

Mohamad Dib

Wireless Networking Engineer

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PROFESSIONAL SUMMARY

Communication Systems Engineer (M.Sc., KTH Royal Institute of Technology) with a focus on wireless networking and the intersection of hardware and software — from embedded firmware and secure wireless communication design to full-stack web development and enterprise network security. Brings hands-on engineering discipline across academic research, IT operations, and production software, with proven adaptability across international environments.

PROFESSIONAL EXPERIENCE

Full-Stack Web Development Intern - The Digital Hub, UNRWA | Beirut, Hybrid

Jun 2026 – Sep 2026

- Built type-safe, production-ready frontends (React, Next.js, TypeScript) and scalable REST APIs (Node.js, Express.js, FastAPI, Django REST Framework).
- Implemented JWT authentication, RBAC, and secure session handling; designed relational and NoSQL (MongoDB) schemas for core app data.
- Integrated LLM-powered features via AI APIs and deployed applications on Vercel/Render using Agile/Scrum and CI/CD workflows.

Field Operations Coordinator - UNRWA / KfW | Siblin Training Centre, Part-Time

Mar 2026 – Apr 2026

- Coordinated field operations independently, meeting UNRWA/KfW quality standards with minimal supervision.
- Maintained accurate attendance, progress, and field documentation to international donor standards.

IT Systems & Technical Support Engineer - Khateeb Home Appliances | Beirut, On-site

Dec 2025 – May 2026

- Administered IT networks and POS systems, maintaining zero downtime via proactive monitoring and troubleshooting.
- Diagnosed and resolved hardware, software, and connectivity issues independently, minimizing operational disruption.
- Optimized inventory database systems, improving data accuracy and retrieval speed for daily operations.

PROJECTS

RestoManager: Full-Stack Restaurant Ordering Platform | [GitHub](#)

- Built the full customer ordering flow (discovery, menu, cart, checkout, live tracking) in React and Node/Express within a 3-person team.
- Contributed admin-side review and approval features supporting the restaurant onboarding workflow.

M.Sc. Thesis: Secure Over-the-Air Computation | [GitHub](#)

- Designed a secure Over-the-Air Computation framework using uniform input distributions and zero-forced artificial noise.
- Built Python/MATLAB simulations of RF impairments and multi-node wireless channels; independently defended thesis to an academic panel.

Vehicular Communication Security (VANETs)

- Engineered frameworks securing VANET communications, focusing on authentication, privacy, & robust security protocols.
- Implemented Public Key Infrastructure (PKI) cryptography to secure data exchange over wireless channels.

TECHNICAL SKILLS

Wireless & RF Engineering: 5G NR / LTE / 3GPP, RF Planning & Analysis, MIMO & Beamforming, Signal Processing (MATLAB)

Networking & Security: TCP/IP, OSPF/BGP, VPNs, Cisco & Fortinet, Zero Trust, Wireshark

Frontend: HTML5, CSS3, JavaScript (ES6+), TypeScript, React.js, Next.js, Zustand, TanStack Query

Backend: Node.js, Express.js, Python, FastAPI, Django, RESTful APIs

Databases & Auth: MySQL, MongoDB, Mongoose, SQL/NoSQL, JWT, RBAC, Secure Cookies

AI & Testing: LLMs, RAG, Prompt Engineering, AI API Integration, Jest, Vitest, Unit/Integration Testing

Architecture & DevOps: Clean Architecture, SOLID, Design Patterns, Git/GitHub, CI/CD, Cloud Deployment

Web & SEO: SSR, CSR, SSG, Core Web Vitals, SEO

EDUCATION

M.Sc., Electrical Engineering, Communication Systems (Wireless Networking Track)

Jan 2025

KTH Royal Institute of Technology, Stockholm, Sweden

B.Sc., Communication Engineering

July 2021

Lebanese International University, Beirut, Lebanon

LANGUAGES

Arabic (Native) | English (Fluent) | Swedish (Conversational)